

CM5235 - Homework Gaussian (2026)

Exercise 1 - Study of the ground state HCl diatomic molecule with different computational methods

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Single-point RHF/STO-3G at $R(\text{H-Cl}) = 1.275 \text{ \AA}$

SCF energy: $E(\text{RHF}) = -455.134835$ Hartree

Electric dipole moment: 1.7343 Debye (components $X = 0$, $Y = 0$, $Z = -1.7343$; along the bond axis)

Molecular orbitals: 10 in total = 9 occupied + 1 virtual. Orbital energies (Hartree):

MO	1	2	3	4	5	6	7	8 (HOMO)	9 (HOMO)	10 (LUMO)
E / Ha	-103.72252	-10.40401	-7.84887	-7.84216	-7.84216	-1.04362	-0.56448	-0.42315	-0.42315	+0.40398

Mulliken charges:

Atom	Cl	H
Charge	-0.171761	+0.171761

1) Optimisation + frequency with six basis sets

Route for each: #p hf/<basis> opt=tight freq.

Comparison table

Basis set	Optimal bond length / Å	Vib. frequency / cm ⁻¹	Number of orbitals	Number of basis functions	Total energy / Hartree	HOMO-LUMO gap / eV
STO-3G	1.3133	3372.5	10	10	-455.136012	21.72
6-31G	1.2953	2924.8	15	15	-460.037173	17.21
cc-pVDZ	1.2773	3149.2	23	23	-460.089450	16.87
6-311+G(d)	1.2714	3106.6	33	33	-460.087170	15.48
cc-pVTZ	1.2673	3138.3	48	48	-460.106851	16.37
cc-pVQZ	1.2656	3138.7	89	89	-460.111272	15.89
Experiment (NIST)	1.2746	2990.9 (ω_e)	-	-	-	-

Comparison with experimental values

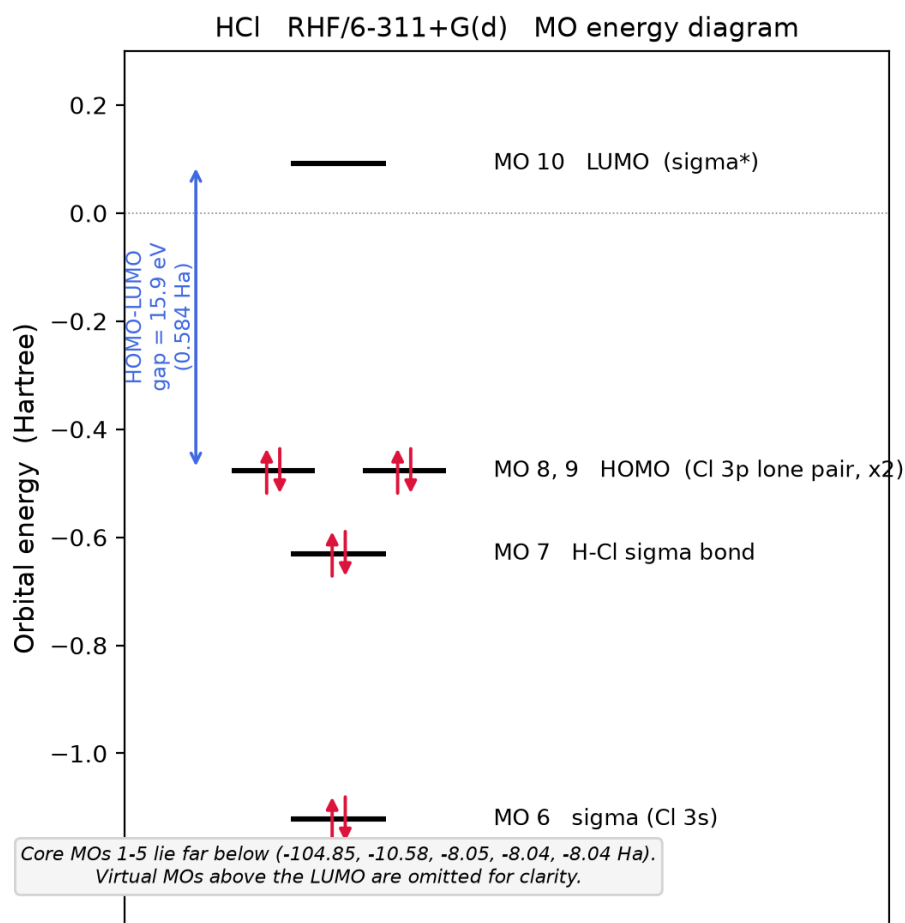
Experimental data from NIST CCCBDB (cccbdb.nist.gov/exp2x.asp, species HCl).

Quantity	STO-3G	6-31G	cc-pVDZ	6-311+G(d)	cc-pVTZ	cc-pVQZ	Experiment
Bond length R_e / Å	1.3133	1.2953	1.2773	1.2714	1.2673	1.2656	1.2746
Vibrational frequency / cm ⁻¹	3372.5	2924.8	3149.2	3106.6	3138.3	3138.7	2990.9
Moment of inertia / 1e-47 kg m ²	2.806	2.729	2.654	2.629	2.612	2.605	2.642
Zero-point energy / kJ mol ⁻¹	20.17	17.49	18.83	18.58	18.77	18.78	17.73
Enthalpy H(298)-H(0) / kJ mol ⁻¹	8.68	8.68	8.68	8.68	8.68	8.68	8.64
Entropy S(298) / cal mol ⁻¹ K ⁻¹	44.686	44.631	44.576	44.557	44.544	44.539	44.67

The calculated bond length, vibrational frequency and moment of inertia converge as the basis set is enlarged; at cc-pVQZ they differ from experiment by -0.009 Å, +148 cm⁻¹ and -0.037e-47 kg m² respectively, whereas the enthalpy H(298)-H(0) and the entropy agree with experiment to within 1%.

2) Orbital energy diagram and occupied MOs (RHF/6-311+G(d))

Orbital energy diagram



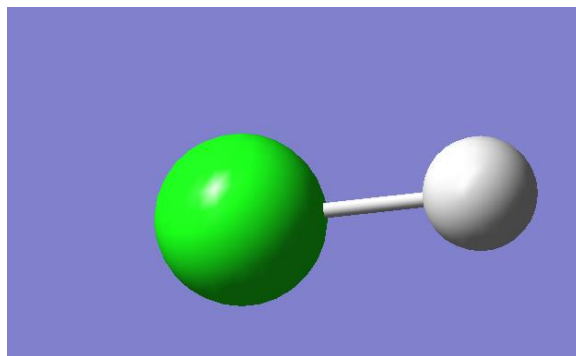
Orbital energy diagram of HCl, RHF/6-311+G(d). Core MOs 1-5 (-104.85 to -8.04 Ha) are off scale; virtual MOs above the LUMO are not shown.

MO	1	2	3	4	5	6	7	8 (HOMO)	9 (HOMO)	10 (LUMO)
E / Ha	-104.84972	-10.57650	-8.04538	-8.04148	-8.04148	-1.12190	-0.63065	-0.47730	-0.47730	+0.09172

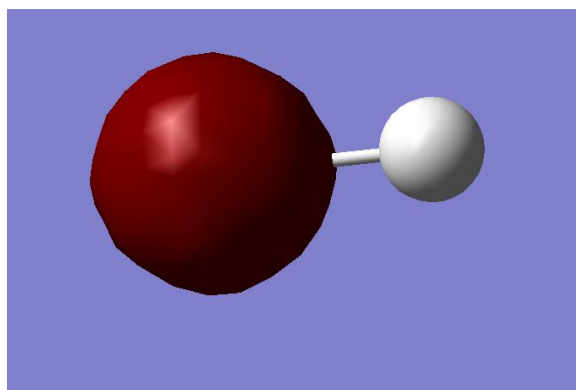
HOMO-LUMO gap = 0.09172 - (-0.47730) = 0.56902 Ha = 15.48 eV.

Occupied molecular orbitals (MO 1-9)

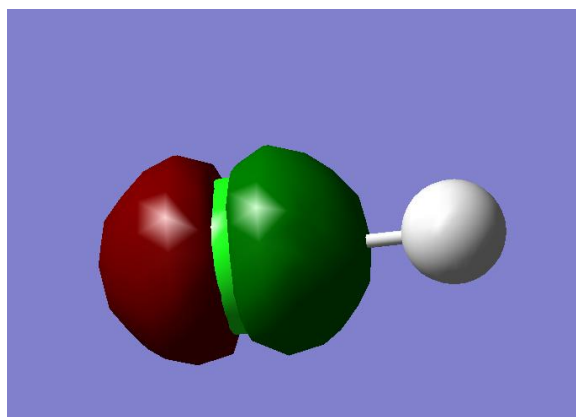
GaussView, isosurface value 0.02; red / green = the two phases of the wavefunction.



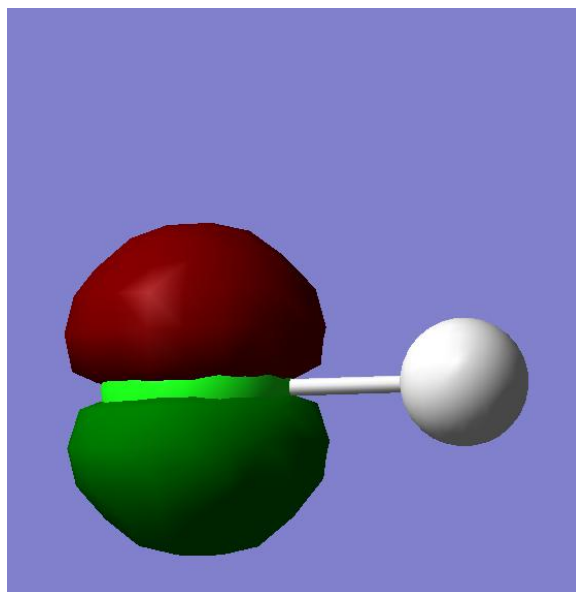
MO 1 ($E = -104.85$ Ha): Cl 1s core (too contracted to show an isosurface at 0.02).



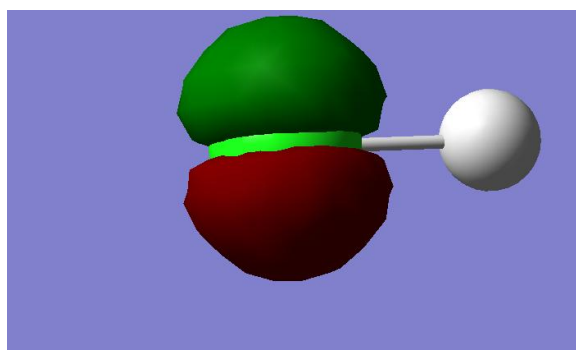
MO 2 ($E = -10.58$ Ha): Cl 2s core.



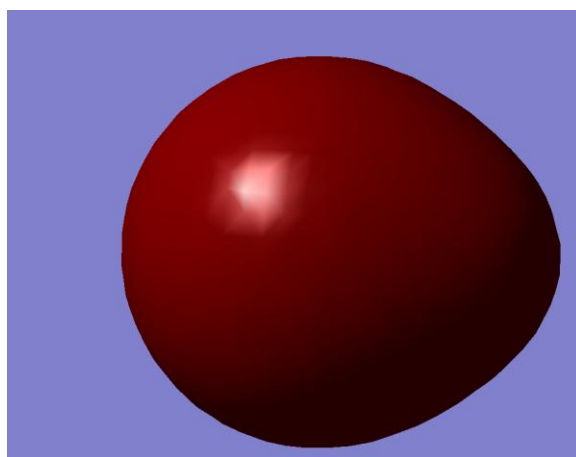
MO 3 ($E = -8.05$ Ha): Cl 2p core (along the bond axis).



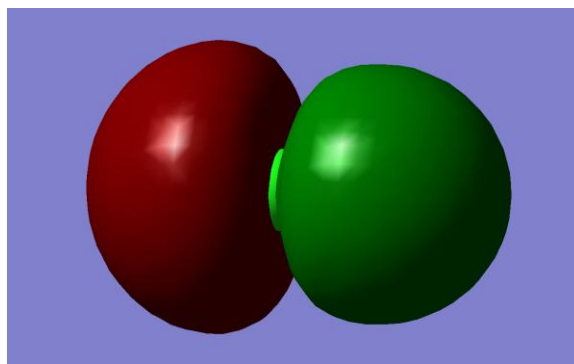
MO 4 ($E = -8.04$ Ha): Cl 2p core.



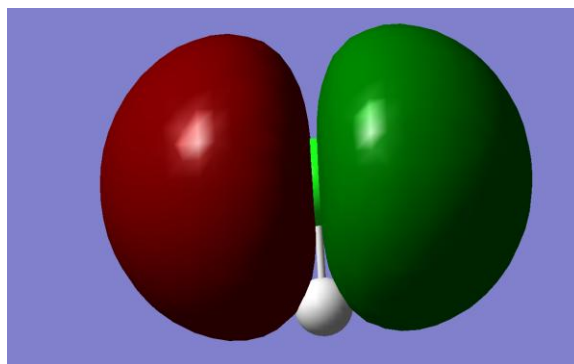
MO 5 ($E = -8.04$ Ha): Cl 2p core (degenerate with MO 4).



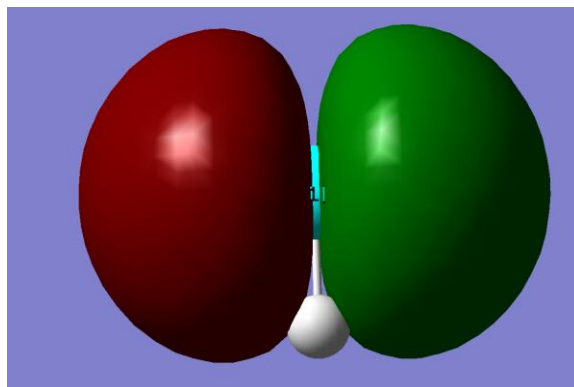
MO 6 ($E = -1.12$ Ha): sigma, dominated by Cl 3s.



MO 7 ($E = -0.63$ Ha): H-Cl sigma bond ($Cl\ 3p_z + H\ 1s$).



MO 8 ($E = -0.48$ Ha): HOMO - Cl 3p lone pair, perpendicular to the bond.



MO 9 ($E = -0.48$ Ha): HOMO - Cl 3p lone pair, degenerate with MO 8.